

SHIVAJI RAO (PhD, MBA, MSC)

DATA SCIENTIST

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Technical Team Leadership | Strategic Roadmap Execution | Client Engagement | Solution Development

Design, Architect and implement AI/ML solutions aligned with market trends and client priorities. Lead and mentor crossfunctional teams, fostering collaboration and innovation. Integrate AI/ML capabilities into existing systems for successful project delivery and client satisfaction.

- Developed AI system to analyze IT projects' impact on regulatory reporting and identify at-risk projects.
- Delivered calculated product delivery windows and optimized logistics operations for improved efficiency and client satisfaction by pioneering large-scale Azure ML system.
- Created Gen AI POC of File based Q & A RAG Chatbot App with LLM (ChatGPT), LangChain, ChromaDB and StreamLit.
- Generated revenue/client value for London-based hedge fund by developing/implementing unsupervised ML model.
- Created a Long/Short Fixed Income Trading model "DRIFT" to assist in making informed critical investment decisions for State Street Bank and Brookfield Asset Management.
- Achieved \$8M in annual cost savings and improved accuracy by 20% through advanced ML techniques at Infosys Consulting.

AREAS OF EXPERTISE

Strategic Planning | Full Life Cycle Project Management | Data Analytics & Visualization | Business Intelligence
Data Warehousing | Agile & Scrum Methodologies | Machine Learning | Staff Training & Leadership | Risk Assessment
Risk Mitigation | Algorithm Development & Implementation | Business Process Automation | Stakeholder Engagement

TECHNICAL BACKGROUND

Languages: Python, C, SQL | **Tools:** FINCAD toolkit, Murex and Numerix toolkit, MongoDB, Django/Nginx/Gunicorn, Bitbucket, GitHub | **Business Reporting:** Tableau | **Software:** ThinkOrSwim, MySQL, RAG, ChatGPT,ChromaDB and StreamLit
| **Cloud Technology:** AWS, MS Azure

PROFESSIONAL EXPERIENCE

BANK OF AMERICA | JERSEY CITY, NJ | MAY 2023–PRESENT SENIOR AI ARCHITECT

Design, Architect and Lead development and deployment of advanced AI system that predicts the impact of IT projects on regulatory reporting, working within international team structures with a focus on innovative ML applications and crossfunctional collaboration.

- Designed using **Random Forest Algorithm to identify potential project failures**, enhancing proactive risk management by **pinpointing top 10 most troublesome projects**.
- Facilitated critical data flows between teams in India and Charlotte to ensure algorithm accuracy and effective ML implementation.
- Implemented ML algorithms using proprietary Phoenix Data Science Platform using Python.
- Created Gen AI POC of File based Q & A RAG Chatbot App with LLM (ChatGPT), LangChain, ChromaDB and StreamLit

INFOSYS CONSULTING | NEW YORK, NY | MAY 2021–MAY 2023 SENIOR PRINCIPAL DATA SCIENTIST

Develop and deploy a large-scale system for a **large logistics firm (FedEx)** in order to calculate the expected date of delivery (EDD) and expected daytime window (EDTW) for each product, with a focus on delivery within the scheduled time. Evaluate the estimates' confidence score, data drift, and model drift for statistical and ML models to determine if more calibration is needed. Provide strategic assistance and design the data and analytics roadmap by identifying and prioritizing business requirements. Develop strategies to produce representative samples for Training and Testing data sets for Machine Learning models. Provide regular communications regarding project status to key stakeholders for informed decision-making.

- Achieved **\$8M** in annual cost savings and improved accuracy by 20% through advanced ML techniques.
- Achieved throughput of **25000 events/second** of processing transactions for the logistics company with each event being a new calculation of EDD and EDTW for the package. Earned award for determining delivery dates and windows for client, **processing 25K transactions per second**.
- Expanded data science and engineering talent bench strength by hiring 10-12 team members.

DATA SCIENCE INSTITUTE, SAINT PETER'S UNIVERSITY | JERSEY CITY, NJ | SEPTEMBER 2019–JULY 2020 ASSISTANT DIRECTOR - DATA SCIENCE

Delivered sessions on **ML techniques** to impact financial decisions and ensure smooth execution of processes. Led 45 students in data science and managed 4-5 professors, including hiring and performance evaluation. Managed master's program's adjunct professors, assigned classes, and oversaw annual budgets.

- Utilized Python to identify optimal locations for **Starbucks openings in New York, Chicago, and Philadelphia**.
- Forecasted bond/stock allocations for individuals aged 60+ to safeguard investments using Python.
- Authored "**On the Application of Machine learning Algorithms to Goals Based Wealth Management (GBWM) with Glide Path Allocations (GPA)**" that was published in the Academy of Business Research.

FAIR VALUE MANAGEMENT, LLC | JERSEY CITY, NJ | OCTOBER 2013–MAY 2021 DIRECTOR, DATA SCIENCE

Employed and managed a large team of over **20 quantitative programmers** to develop investment applications leveraging the FINCAD toolkit, as well as knowledge acquired with the Murex and Numerix toolkits. Utilized Python 2.7.x/3.5.6, R, MySQL, MongoDB, Django/Nginx, and Unicorn on AWS to design, build, and deploy all investment apps while coordinating and managing changes worldwide with Git. Supervised a group of interns (Stevens Institute and Stony Brook University) in Machine Learning research utilizing Keras TensorFlow and Tableau data visualization. **Managed multimillion-dollar budget/hired all team members.**

- Produced \$500K in revenue by developing Long/Short Equity Trading Model for London-based Hedge Fund using unsupervised ML.
- Generated \$1M+ in revenue by creating Long/Short Fixed Income Trading model, "**DRIFT**," to assist in making informed investment decisions for **State Street Bank** and **Brookfield Asset Management**.

VANTAGE CAPITAL MARKETS | LONDON, ENGLAND | JULY 2012–SEPTEMBER 2013 CO-HEAD OF CRE DISPOSITIONS

Developed investment platform to calculate the IRR for distressed residential and commercial real estate, wind farm, and transportation loans. Identified business opportunities and matched US buyers with European sellers. Managed and hired 3 team members.

- Enabled **York Investments Limited** to acquire **€115M warehouse portfolio** by providing distressed asset project data.
- Secured business deal with IRR of 19%, demonstrating a high return on investment.

MF GLOBAL | NEW YORK, NY | NOVEMBER 2009–MARCH 2012 SENIOR VICE PRESIDENT, FIXED INCOME

Developed several Visual Basic and C# applications for fixed income trading strategies. developed Rich and Cheap calculator programs for On the Run/Off the Run/Off-Off-the-Run Treasury Securities. Created Futures Treasury Roll Calculator to determine the expected Roll, an Auction Model to trade the US Treasury Auctions of the 2year/5year/7year and the 3year/10year/30year auction cycle during the month, and a Fed Fund Calculator to determine the fair value of the Fed Fund Strip, the Euro Dollar Strip, the Options on the Fed Fund and the Options on Eurodollar Futures.

Key Contributions:

- Designed many Visual Basic and C# programs for trading methods in fixed income. Developed Rich and Cheap calculator for On the Run/Off the Run/Off-Off-the-Run Treasury Securities.
- Created a Futures Treasury Roll Calculator to determine the expected Roll, an Auction Model to trade the 2year/5year/7year and 3year/10year/30year auction cycles during the month, and a Fed Fund Calculator to determine the fair value of the Fed Fund Strip, the Euro Dollar Strip, the Options on the Fed Fund, and the Options on the Eurodollar Futures.
- Established a relative value program to decide the bonds to be purchased and sold during buybacks and sellbacks (Quantitative Easing and Reverse QE) of Treasury bonds coordinated by the Federal Reserve and the Department of the Treasury. Designed a scheme to benefit from the mispricing in the US Treasury market brought on by the financial crisis of 2008.
- Generated ~\$20M in customer fees by executing transactions on behalf of clients between 2009 and 2012.

Additional Experience:

Adjunct Assistant Professor, Stevens Institute of Technology, Hoboken, NJ

Managing Director & Co-Founder, Global Macro Group at Cantor Fitzgerald & Co, New York, NY

EDUCATION

Doctor of Philosophy (Ph.D.), Operations Research, Baruch College, The City University of New York, New York, NY

Master of Science (M.Sc.), Management Science, Durham University, Durham, UK

Bachelor of Arts (B.A.) with Honors in Economics and Mathematics, Shri Ram College of Commerce, New Delhi, India

CONTINUOUS LEARNING

AI Prompting by New Jersey Institute of Technology, Feb 2025.

Gen AI Pinnacle Program by Analytics Vidhya 2024

Data Science Immersive, General Assembly, March 2021

Python for Data Bootcamp, General Assembly, 2020

Introduction to Agile and Scrum, General Assembly, 2020